An Equivalent Circuit Model for 2D Embroidery Using Conductive Threads

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Abstract

2D embroidery techniques, such as fill stitch, enable the creation of conductive areas in e-textiles. Predicting their electrical resistance is however difficult due to dense rows simultaneously lengthening current paths and creating numerous inter-thread contacts. We present a physics-based equivalent circuit model that links these competing effects by modeling each inter-thread junction with a through-thread resistor in parallel with a contact resistor. Our method estimates fill stitch resistance from key parameters including thread resistivity and radius, fabric thickness, and stitch density. Evaluations with embroideries using 33 combinations of stitch densities, embroidery sizes, and shapes show that experimental results closely match our model's predictions. Our work provides a quantitative foundation for understanding and systematically designing 2D embroidered smart textiles.

CCS Concepts

• **Hardware → Sensor devices and platforms**; *Flexible and printable circuits*; • **Human-centered computing → Interaction devices**; *Ubiquitous and mobile computing systems and tools.*

Keywords

Interactive embroidery, conductive thread, smart textile.

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1 Introduction

Computer-controlled embroidery places threads with 0.1mm precision, streamlining the process of embedding sensors and actuators directly into fabric. This approach opens up possibilities for a wide range of applications, from wearable health monitoring [11] and interactive art [3] to smart garments [1, 10].

However, a fundamental challenge hinders the reliable performance of embroidered e-textiles: the inherent and often unpredictable variability of electrical resistance in conductive threads once stitched into fabric. Unlike rigid electronic components with well-defined electrical properties, the resistance of an embroidered conductive thread fluctuates considerably. This fluctuation results from several design and material factors, including stitch type (e.g. a simple line versus a filled area), stitch parameters (e.g. stitching length and density), and properties of the conductive thread itself (see Figure 1 for an overview of fundamental embroidery techniques and relevant terminology). This resistance inconsistency can lead to difficulties in sensor calibration, inefficient power consumption, and potentially complete malfunction of the intended device. Accurate estimation of this resistance *prior* to fabrication is therefore critical for the successful development of embroidered e-textiles.

While previous research has proposed equivalent circuit models to estimate the resistance of conductive threads [9, 12], these models predominantly focused on one-dimensional (1D) embroidery, such as the running stitch technique (Figure 1-b) commonly used for wiring. These models often do not adequately account for the complexities arising in two-dimensional (2D) embroidery, such as fill stitches (Figure 1-b). This is a significant limitation as fill stitches are essential for creating planar conductive areas that serve as electrodes for sensors or other interactive elements. In such dense 2D structures, the repeated contact between adjacent conductive threads introduces a substantial contact resistance, a factor largely neglected by existing 1D models.

In this note, we address this shortcoming by developing a physics-based equivalent-circuit model specifically for conductive fill stitches in 2D embroidery. Our model accounts for both the intrinsic resistance of the conductive thread segments and, critically, the contact resistance at the junctions between adjacent threads. We focus on the fill stitch because of its prevalence in embroidery applications and its characteristic dense packing of threads, which

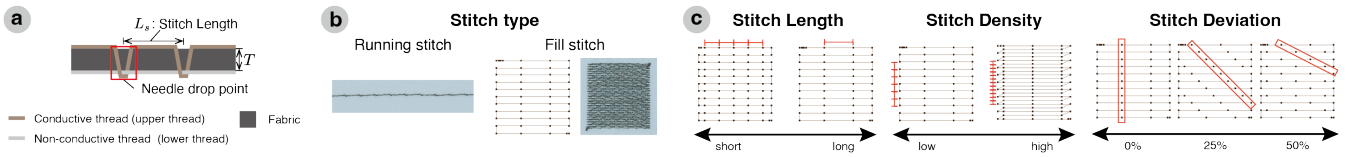


Figure 1: Embroidery techniques and parameters. (a) Cross-section of a single embroidered stitch. An upper thread (e.g. conductive) interlocks with a bobbin thread below the fabric. L_s denotes stitch length and T fabric thickness. (b) Stitch types. A *running stitch* forms a 1D trace for wiring, while a *fill stitch* (or *tatami stitch*) comprises close parallel stitch rows for 2D areas like electrodes. (c) Key stitch properties: *Stitch Length* is the upper thread length between needle drops; *Stitch Density* is the spacing between rows in a fill stitch affecting inter-thread contact, and *Stitch Deviation* (or *Stagger/Offset*) refers to needle drop displacement, affecting texture and interaction. Stitch density in fill stitches significantly impacts the electrical properties (e.g. resistance) of conductive thread patterns due to path length and inter-thread contact.

accentuates the effect of contact resistance that previous models overlook.

In summary, our contributions are:

- A novel equivalent circuit model for 2D embroidered patterns that accurately integrates both through-thread and inter-thread contact resistance.
- Experimental validation of the model's predictive power using embroidered deformation sensors, whose electrical response is directly governed by variations in stitch density.

2 Related Work

Our work builds on two areas of prior research: the use of embroidery as a fabrication method for smart textiles, and the development of models to predict the electrical properties of the resulting structures.

2.1 Embroidery for smart textile fabrication

While traditionally a decorative art, machine embroidery has emerged as a powerful technique for fabricating smart textiles. By substituting standard thread with conductive thread, intricate and detailed electrode patterns can be created for a wide range of interactive devices [3, 5, 7, 8]. This method allows designers to directly engineer the electrical properties and sensing capabilities of a textile by adjusting stitch parameters such as stitch density and length. For instance, Shinoda et al. created capacitive touch sensors by overlaying embroidered electrodes with non-conductive thread, showing that varying the stitch density of the overlay could produce distinct capacitive regions for touch sensing [10]. Similarly, Aigner et al. evaluated how stitch length and count affect the dynamic range of embroidered pressure sensors [1].

These studies successfully demonstrate that tuning embroidery parameters can yield unique sensing capabilities. However, they rely on empirical exploration for a limited set of patterns and lack a generalizable, predictive model to systematically guide the design process. Our work aims to provide this theoretical framework, focusing on how stitch properties fundamentally determine electrical resistance.

2.2 Embroidery design parameters and their electrical properties

For the reliable design of embroidered electronics, predicting their electrical behavior before fabrication is critical. Investigating the influence of different parameters to create conductive paths, Akgün found that thread type, stitch density, and thread position were significant contributors to the resulting electrical resistance [2]. Juchnevičienė et al. examined how stitch density and the width of a conductive trace influence DC resistance, finding that denser and wider patterns exhibit lower resistance [6]. Their work also revealed complex, non-linear interactions, such as a strong correlation between resistance and fabric properties appearing only at high stitch densities.

Building on these empirical insights, researchers have developed theoretical models to anticipate resistance prior to fabrication. For instance, Zhang et al. proposed geometric and equivalent resistance circuit models (Figure 1-a) [12] using stitch length and density as parameters. Likewise, Rho et al. created a model to predict both thread consumption and resistance for running and zigzag stitches, based on parameters like stitch length, direction and layering [9].

While useful, these modeling efforts share a critical limitation: they are designed for 1D embroidery patterns. They do not account for the complex electrical pathways that arise in 2D structures created with a fill stitch. In these dense, planar areas, which are essential for components like sensor electrodes, the electrical current flows not only through the length of the thread but also across adjacent, contacting threads. This inter-thread contact resistance is a significant factor in the overall conductivity of 2D patterns but is neglected in existing 1D models. Our work directly addresses this gap by developing an equivalent circuit model specifically for 2D fill stitches that explicitly integrates the effects of inter-thread contact resistance, providing a more accurate and comprehensive predictive tool for designers.

3 Equivalent Circuit Model

3.1 Model Derivation for a Running Stitch

A running stitch (Figure 1-a) forms a linear conductive path. At each needle drop, the upper conductive thread passes through the fabric and interlocks with the bobbin thread. For a surface path of length L and stitch length L_s, the consumed upper-thread length is:

$$L_t = L + 2T \left\lfloor \frac{L}{L_s} \right\rfloor$$

where $\lfloor L/L_s \rfloor$ is the number of times the thread passes through the fabric. Resistance scales with L_t and the thread's resistivity.

3.2 Model Derivation for a Fill Stitch

We now extend the principles from the running stitch model to develop a model for the fill stitch. A fill stitch can be conceptualized as a series of dense, meandering running stitches that form a 2D conductive area (Figure 1-b). However, this dense packing introduces a critical new factor: mutual electrical contact between adjacent conductive thread segments. These contacts create additional pathways for current, significantly influencing the overall resistance.

Modeling every precise point of mutual contact in a flexible, nonuniform embroidered structure is computationally intractable. We therefore introduce a necessary simplification: we treat each conductive thread segment as an ideal cylinder of radius r. Figure 2 provides a visual guide for the concepts discussed in this section. The distance between the centers of two adjacent parallel thread segments is denoted by d (Figure 2-a).

A close-up view of the threads in a fill stitch reveals that they are not perfectly parallel or straight. The embroidery process, where the machine feeds successive stitches, introduces a slight angle, resulting in a saw-tooth pattern of threads (Figure 2-d). This inherent angular placement impacts the proximity and contact points between adjacent conductive threads. To capture this, we calculate the effective center-to-center spacing, $d_{\text{diff}}(x)$, between two neighboring conductive threads along their length x using the following piecewise formulation:

$$d_{\text{diff}}(x) = \begin{cases} d_{\text{design}} + \frac{4r}{L_s}x + 2r(D_{\text{dev}}-1), & x_i < x \le (1-D_{\text{dev}})x_{i+1},\ x_i - x_{i+1} < 0 \\ d_{\text{design}} + \frac{4r}{L_s}x + 2r(D_{\text{dev}}-2), & (1-D_{\text{dev}})x_{i+1} < x < x_{i+1},\ x_i - x_{i+1} < 0 \\ d_{\text{design}} - \frac{4r}{L_s}x + 2r(1-D_{\text{dev}}), & x_{i+1} < x \le (1-D_{\text{dev}})x_i,\ x_i - x_{i+1} > 0 \\ d_{\text{design}} - \frac{4r}{L_s}x + 2r(2-D_{\text{dev}}), & (1-D_{\text{dev}})x_i < x < x_i,\ x_i - x_{i+1} > 0 \end{cases}$$

where x_i is the x-coordinate of the i-th needle drop point along the stitch direction, d_{design} is the nominal inter-row spacing specified in the embroidery design file (related to stitch density), L_s is the stitch length, and D_{dev} (with $0 \le D_{\text{dev}} < 1$) is the embroidery deviation or stagger parameter (Figures 1-c and 2-d). This formulation captures how the angled path alters the thread gap. The saw-tooth pattern depends on the sewing-machine thread-tension settings. Following standard embroidery practice, we adjust the tension so that the lower thread remains straight and taut on the fabric's underside, where it stays out of view. Under this condition, the upper thread loops around the lower thread—emerging at the surface, passing to the back, and returning to the front—thereby forming the saw-tooth geometry illustrated in Figure 2-d.

While $d_{\text{diff}}(x)$ can theoretically be very small, in reality, threads cannot infinitely compress. One thread will typically ride over another. We model this by defining a minimum practical separation distance, $d_{\min}$, as a fraction α_d (an empirical parameter) of the thread radius r. Thus, the actual separation d used in our model is:

$$d = \max(d_{\text{diff}}(x), \alpha_d r)$$

When adjacent threads come into contact (i.e. when $d < 2r$), they form an electrical junction. We model this junction as an additional resistive path. The equivalent circuit for a small segment ΔL of two adjacent threads includes their intrinsic resistance and this contact resistance, R_c (Figure 2-b). The geometry of this contact, viewed as the intersection of two cylinders (Figure 2-c), determines the length of the contact surface, l_c, calculated as:

$$l_c = 2\arccos\left(\frac{d}{2r}\right)$$

For resistance calculation, l_c relates to the effective width of the contact area. Based on Holm's theory of electrical contacts [4], the contact resistance R_c over this infinitesimal area can be defined as:

$$R_c = \frac{\rho}{2l_c\alpha_a}$$

where ρ is the volume resistivity of the conductive thread material, and α_a is another empirical parameter representing the ratio of the actual effective electrical contact area to the apparent geometric contact area. This factor accounts for surface roughness and imperfect contact.

The model also needs to consider the resistance of the conductive thread itself, R_T, along the segment ΔL. If ρ is the linear resistance per unit length of the conductive thread, R_T for a segment ΔL is:

$$R_T = \begin{cases} \rho_L \Delta L & \text{if not at a needle drop point} \\ \rho_L(\Delta L + 2T) & \text{if at a needle drop point (thread passes through fabric)} \end{cases}$$

This formulation adapts the running stitch concept, where the thread path increases by $2T$ at needle drop points due to fabric thickness T.

The overall equivalent circuit for the fill stitch consists of a network of these R_T elements (representing current flow along the threads) and R_c elements (representing current flow between adjacent threads). For a fill stitch of total length L_{fill}, there are $N = L_{\text{fill}}/\Delta L$ such elementary segments in series.

This model is applied across the fill stitch area, with two specific considerations:

- **At needle drop points:** The conductive thread passes through the fabric to interlock with the bobbin thread. At these points, the segment of the thread passing through the fabric does not make side-contact with adjacent parallel threads on the surface. Thus, there is no contact resistance for these specific portions.
- **At turn-around points (edges of embroidery):** At the edges of a fill-stitched area, the thread turns to start a new parallel row. This creates a loop of conductive thread on the underside of the fabric (Figure 2-e). This loop contributes to the overall resistance. We model this turn-around resistance by considering it as a short embroidered segment subject to both through-thread resistance R_T and potential self-contact resistance R_c within the loop. The effective contact proportion within this loop is represented by a third empirical parameter, α_l.

By discretizing the entire embroidered pattern into these ΔL segments and calculating the R_T and R_c values, a comprehensive resistance network can be constructed and solved to estimate the overall electrical resistance of the 2D embroidered structure.

4 Evaluations on Sizes and Shapes

We first evaluated the effect of embroidery size and shape on resistance. The experimental conditions for this initial set of evaluations were as follows: **(i) Varying size (fixed shape):** 20, 28, and 36 mm. Shapes were all squares, **(ii) Varying shape (fixed size):** *Square*, equilateral-*triangle*, and *circle*. Sizes fixed at 28 mm.

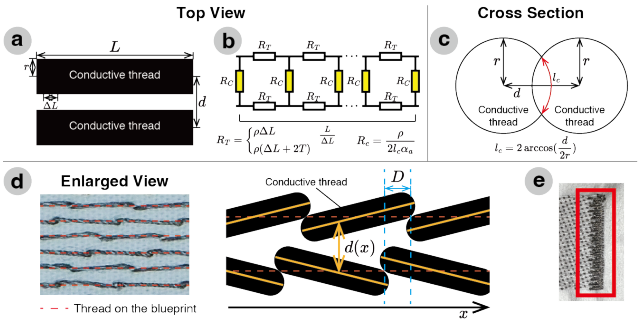


Figure 2: Definition of equivalent resistance circuit model. (a) Top view of a sequence of conductive threads. (b) Top view of equivalent circuit of conductive threads. (c) Cross section of conductive threads in contact. (d) Saw-tooth pattern produced by the embroidery machine, which feeds each successive stitch at a slight angle. (e) The back side of the turned stitches at the edge of the embroidery.

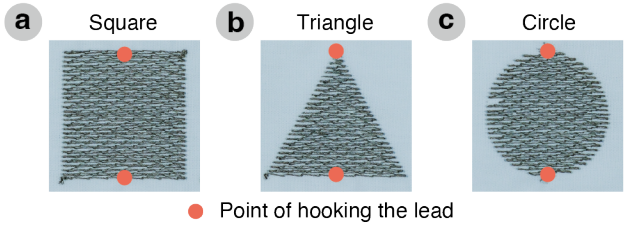


Figure 3: Positions we hooked leads to measure the resistance.

Preliminary tests indicated that resistance values stabilized when stitch density exceeded 3.0 threads/mm. We therefore focused on five stitch densities—1.0, 1.5, 2.0, 2.5, and 3.0 threads/mm—to closely examine the range where significant resistance variations were anticipated. Standard machine-embroidery settings for stitch length (4 mm) and stitch deviation (25%) were used, as our initial checks confirmed these parameters had a limited impact on resistance in this context. We created five samples for each unique condition, resulting in 125 samples in total for this phase[1].

4.1 Apparatus and Materials

Samples were embroidered on a Tajima Sai machine (10 × 10 cm frame) at 250 spm to minimize coating wear [9]. Embroidery patterns were generated from digital data[2] created with custom scripts based on pyembroidery[3] and Stitch Generator[4]. Resistance measurements were taken with a Sanwa Electric Instrument Co., Ltd. CD771 digital multimeter. To ensure consistent contact, a Sanwa TL-9IC clip lead was used to hook only a single thread at designated points for each shape (details in Figure 3). All measurements were conducted after removing the embroidery from the frame, with the fabric unfixed.

We used broadcloth with an adhesive backing (total thickness 0.47 mm). The upper thread was Smart-X (Fujix; 100% nylon, silver-coated, 0.32 mm; nominal 250 Ω/m); the bobbin thread was

[1]Shape and size conditions were tested separately, but the 28 mm square samples, being common to both, were produced only once.
[2]Embroidery data (array of needle drop coordinates) acts as a blueprint for the machine
[3]https://github.com/EmbroidePy/pyembroidery
[4]https://github.com/bastanja/stitch_generator

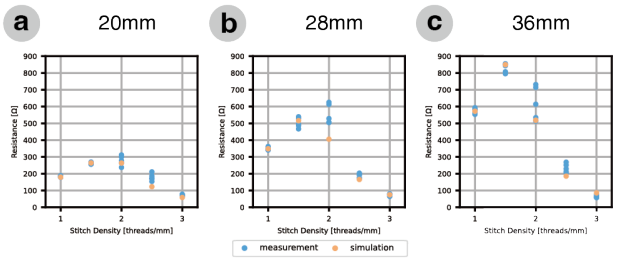


Figure 4: Measured and simulated resistance values for different embroidery sizes. (a) 20mm, (b) 28mm, (c) 36mm.

polyester. To calibrate wear-induced changes, we embroidered 10 cm running stitches (five samples each at 2, 3, and 4 mm stitch lengths), measured resistance over the central 9 cm, extracted the consumed thread length, and computed the effective linear resistance as (measured resistance)/(consumed length), yielding 442 ± 29Ω/m, which we used in all simulations.

4.2 Simulation

Our equivalent-circuit model is initialized with parameters derived from the embroidery design file and basic material measurements:

- Fabric thickness (T_{fabric}).
- Conductive thread radius (r_{thread}) and its effective linear resistance (ρ_L, from Section 4.1).
- Design-specified nominal row spacing (d_{design}), from which stitch density ($1/d_{\text{design}}$) is derived.
- Overall embroidered pattern length (L) and stitch deviation (D).

Since the needle drop points are defined with a precision of 0.1 mm in the embroidery design data, we use a segment length of $\Delta L = 0.1$ mm to discretize the embroidered pattern for our model.

Three empirical coefficients, which are difficult to measure directly due to factors like surface roughness, coating wear, and local thread compression, are required by the model:

- α_d: Defines the minimum effective center-to-center gap between adjacent conductive threads, as a fraction of the thread radius.
- α_a: Represents the ratio of the actual electrically conductive area to the apparent (geometric) contact area between threads.
- α_l: Specifies the effective contact proportion of the thread loop formed on the fabric's reverse side during turn stitches at pattern edges.

These coefficients were estimated using Leave-One-Group-Out Cross-Validation (LOGO-CV). In this process, for each experimental condition, its measured resistance data was temporarily excluded. The coefficients $\alpha_d, \alpha_a, \alpha_l$ were then fitted to the remaining dataset via non-linear least-squares optimization. The resulting parameter set was used to predict the resistance for the excluded condition. Repeating this for all conditions yielded a single, physically plausible set of coefficients and provided an unbiased assessment of the model's predictive accuracy.

4.3 Results

Figure 4 illustrates the resistance trends for square embroideries of three side-lengths (20 mm, 28 mm, and 36 mm) across the tested stitch densities. As anticipated, increasing the side length

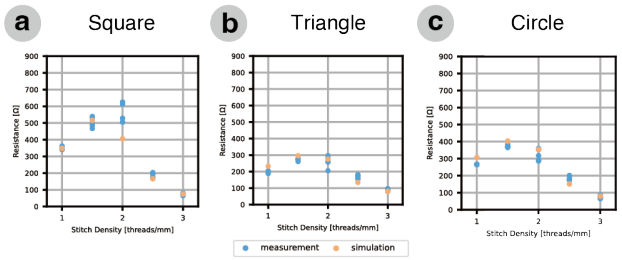


Figure 5: Measured and simulated resistance values for different embroidery shapes. (a) Square, (b) Triangle, (c) Circle.

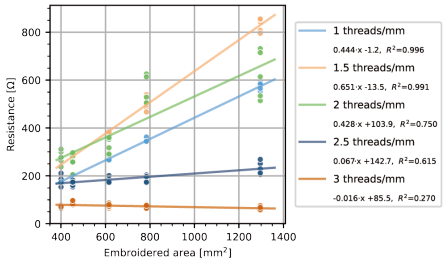


Figure 6: The relationship between embroidery area and resistance for each stitch density. Dots indicate measured values, and the lines represent linear regression models fitted to those measurements. In order of decreasing area: square (20 mm), triangle (28 mm), circle (28 mm), square (28 mm), square (36 mm).

from 20 mm to 36 mm resulted in a higher overall resistance at every stitch density. Importantly, all three sizes exhibited a consistent convex relationship with stitch density: resistance initially increased, peaking around 1.5–2.0 threads/mm, and then decreased sharply as higher densities introduced more parallel paths via inter-thread contacts.

Figure 5 presents a comparison of square, equilateral-triangle, and circular patterns, all with a characteristic dimension of 28 mm. Due to inherent differences in total thread length and inter-thread contact patterns for a given stitch density, the baseline resistances at 1.0 thread/mm varied by shape (circle > square > triangle). Despite these differences, all shapes displayed the same characteristic resistance peak at ≈ 1.5–2.0 threads/mm, followed by a monotonic decrease. This consistent turning point suggests that inter-thread contact density, more than the specific outline geometry, is the primary driver of this non-monotonic behavior.

Quantitatively, the model's predictions showed good agreement with these experimental measurements. For the size variation experiments, the mean-absolute-error (MAE) was 35.0 Ω, the root-mean-square-error (RMSE) was 60.4 Ω, and the mean-absolute-percentage-error (MAPE) was 12.6%. For the shape variation experiments, these metrics were: MAE = 32.4 Ω, RMSE = 49.9 Ω, and MAPE = 12.2%.

4.4 Discussion

Overall, the simulation results from our model align well with the measured resistance changes. As seen in Figures 4 and 5, most simulated data points closely track the experimental measurements, with the mean percentage error generally remaining below 15%.

One notable deviation is observed for the 28 mm square sample at a stitch density of 2 threads/mm, where the simulation underestimates the measured peak resistance. However, given that all other tested sizes and shapes consistently follow the same convex resistance curve, with geometry primarily scaling the absolute resistance values rather than shifting the critical turning point density, this specific discrepancy at 2 threads/mm for the 28 mm square warrants a more focused investigation. This observation informed the design of our subsequent study, where we specifically use the 28 mm square to probe density-related effects more thoroughly while minimizing geometric confounds.

The relationship depicted in Figure 6, showing embroidery area versus resistance for various stitch densities, further elucidates the interplay between through-thread conduction and inter-thread contact resistance. In sparsely stitched samples, conductive threads tend to be electrically isolated. Consequently, increasing the stitched area (which also increases the total length of conductive thread) results in a proportional increase in overall resistance. In contrast, for densely stitched samples, a significant network of inter-thread junctions forms. These junctions create multiple parallel pathways for current, leading to a situation where the overall resistance becomes largely insensitive to further increases in the embroidered area.

5 Evaluations on Stitch Density

A key aspect of our equivalent circuit model is that it takes into account contact resistance (as discussed in Section 3). Therefore, we evaluated the accuracy of our model for different stitch densities, focusing on the following 13 conditions: 1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 2.0, 2.2, 2.4, 2.5, 2.6, 2.8, and 3.0 threads/mm. Our previous evaluation of shape and size revealed that a 28 mm square yields the greatest resistance contrast across stitch density conditions while remaining compact enough to avoid secondary issues such as fabric distortion. We therefore adopted this geometry for the stitch density evaluations and tested all densities on 28 mm squares. We used the same apparatus, materials, and protocols to measure the resistance values as in the first evaluation.

5.1 Results

Figure 7 plots the resistance of square fill stitch samples as a function of stitch density. The measured data (blue) exhibit a non-monotonic trend: resistance increases from about 350 Ω at 1.0 thread/mm to a peak of 580–630 Ω around 1.8–2.0 threads/mm, then falls rapidly to < 100 Ω at 3.0 threads/mm. We performed the same quantitative evaluation for stitch density (13 densities × 5 replicates = 65 pairs). With our data, the model achieves an MAE of 32.4 Ω, an RMSE of 53.9 Ω, and a MAPE of 9.8%. Averaging the five samples at each density and comparing the 13 means with the model yields even smaller errors (MAE = 26.1Ω, RMSE = 46.6Ω, MAPE = 8.2%). These results suggest that most remaining discrepancies stem from random differences between samples rather than from the model.

5.2 Discussion

Our model effectively captures resistance variations with stitch density. Resistance shows a convex trend, peaking at around 1.5 to 2.0 threads/mm, then decreases with higher densities. This is due to two mechanisms: at low densities, threads do not contact each

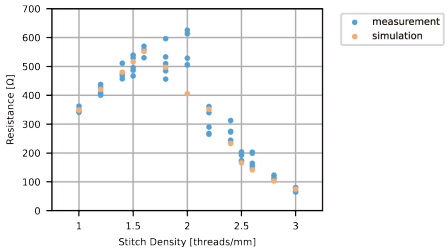


Figure 7: Measured and simulated resistance values for different stitch densities.

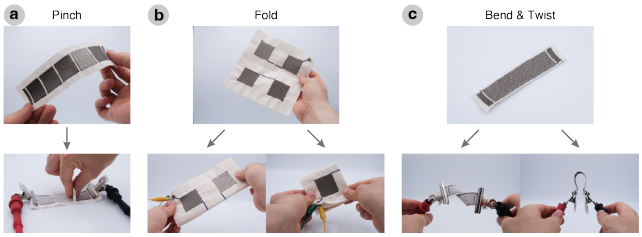


Figure 8: Sensor design examples. (a) a pinch sensor, (b) a fold-aware cloth, and (c) a bending/twisting sensor.

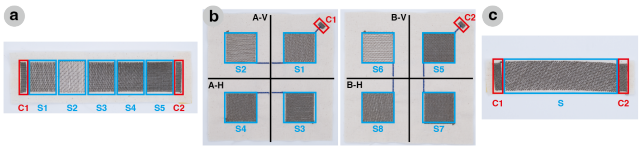


Figure 9: Sensor electrode arrangement. (a) Pinch Sensing, (b) Fold Sensing, (c) Bend/Twist Sensing.

other, causing longer conductive paths and increased resistance, aligning with 1D embroidery models [9, 12]. At higher densities, mutual thread contacts form low-resistance junctions, reducing overall resistance. Our circuit model includes both conduction and junction resistances, accurately reflecting the resistance curve.

The only conspicuous gap appears in the narrow band around 2 threads/mm, where it is difficult to predict whether neighboring conductive threads are in contact. A small difference in other factors, such as the diameter of the thread and the thickness of the fabric, may determine the likelihood of occurrence of mutual thread contacts. The measurements mirror this sensitivity: the variances at 1.8-2.2 threads/mm are wider than at other densities. Our examination suggests that designers should avoid this stitch density range when tight resistance tolerances are required.

6 Sensor Design Demonstrations

To demonstrate the utility of the model, we developed sensors (Figures 8 and 9) that detect finger pinches, folding shapes, and bending/twisting directions using stitched electrodes on fabric. For pinch detection, five square electrodes with varying stitch densities (2.0, 1.0, 2.4, 2.8, and 4.0 threads/mm) create distinct resistance profiles. Pinching alters resistance between contacts, enabling precise location detection with a micro-F1 of 1.00 in our 5-folded cross validation. For fold detection, eight electrodes with 1.0, 1.5, 2.0, 2.3, 2.5, 2.8, 3.0, and 4.0 threads/mm in a specific sequence identify 13 fold configurations with a micro-F1 of 0.938 across classes, exhibiting distinct resistance levels for most configurations. Bend and twist direction detection is achieved with a single bar electrode; varying stitch density (1.5, 2.0, 2.5, and 3.0 threads/mm) improves sensitivity, with a micro-F1 of 0.933 for optimal density (2.0 threads/mm). The sensors operate efficiently using two electrical connections. Previous pinch- and fold-sensors require a separate trace and cloth-to-PCB connection for every electrode. By assigning each serially linked electrode a distinct stitch density—and therefore a distinct baseline resistance—we discriminate events through the two pads at the chain's ends alone. The same density design lets a single strip sense bending and twisting. This approach is feasible only because our model predicts how stitch density shifts resistance.

7 Limitations and Future Work

We discuss several limitations to clarify the contributions and future research directions of our work. An 8-needle industrial embroidery machine minimizes uncertainty from conductive thread wear, but results may vary with a single-needle household machine due to different thread paths, tensions, and speeds affecting resistivity. Our method adapts by calibrating resistance with running-stitch samples, ensuring the simulation accounts for machine-specific abrasion effects. The model focuses on stitch density over material, allowing sensor fabrication with a single-needle machine.

Similarly, different threads and cloths may lead to different results. We have not demonstrated the generalizability of our model across different materials, but we believe it serves as a foundation for further investigations of the electrical properties of other embroidery structures.

Furthermore, our current model only covers the stitch density of fill stitches, and future work should investigate the effects of other factors, such as different stitch types (e.g. cross-stitch), stitch length, stitch deviation, etc. to form a more general model.

8 Conclusion

We developed an equivalent-circuit model that captures both linear and contact resistance in two-dimensional fill stitching. Using only design-time parameters, such as thread resistivity, stitch density, and fabric thickness, our model accurately predicts embroidery resistance. We demonstrated its practical utility by implementing a deformation sensor that utilizes stitch density variation as the basis of its response.

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References

[1] Roland Aigner, Andreas Pointner, Thomas Preindl, Patrick Parzer, and Michael Haller. 2020. *Embroidered Resistive Pressure Sensors: A Novel Approach for Textile*

Interfaces. Association for Computing Machinery, New York, NY, USA, 1–13. https://doi.org/10.1145/3313831.3376305

[2] Duygu Erdem Akgün. 2022. Effecting factors on electrical resistance of conductive paths. *Journal of Industrial Textiles* 51, 2_suppl (2022), 3004S–3013S. doi:10.1177/15280837221077704

[3] Nur Al-huda Hamdan, Simon Voelker, and Jan Borchers. 2018. Sketch Stitch: Interactive Embroidery for E-Textiles. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3173574.3173656

[4] Ragnar Holm. 2013. *Electric Contacts: Theory and Application*. Springer Berlin Heidelberg. https://books.google.co.jp/books?id=yfzuCAAAQBAJ

[5] Jeyeon Jo and Cindy Hsin-Liu Kao. 2021. SkinLace: Freestanding Lace by Machine Embroidery for On-Skin Interface. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI EA '21)*. Association for Computing Machinery, New York, NY, USA, Article 444, 6 pages. doi:10.1145/3411763.3451756

[6] Žaneta JUCHNEVIČIENĖ, Milda JUCIENĖ, and Svetlana RADAVIČIENĖ. 2017. The research on the width of the closed-circuit square-shaped embroidery element. *Materials Science* 23, 2 (2017), 186–190.

[7] Hua Ma and Junichi Yamaoka. 2022. SenSequins: Smart Textile Using 3D Printed Conductive Sequins. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) *(UIST '22)*. Association for Computing Machinery, New York, NY, USA, Article 85, 13 pages. doi:10.1145/3526113.3545688

[8] E. R. Post, M. Orth, P. R. Russo, and N. Gershenfeld. 2000. E-broidery: Design and fabrication of textile-based computing. *IBM Systems Journal* 39, 3.4 (2000), 840–860. doi:10.1147/sj.393.0840

[9] Soo Hyeon Rho, Suhyun Lee, Wonyoung Jeong, et al. 2022. Study of the optimization of embroidery design parameters for the Technical Embroidery Machine: derivation of the correlation between thread consumption and electrical resistance. *Textile Research Journal* 92, 9-10 (2022), 1550–1564.

[10] Kazuhiro Shinoda, D. Antony Chacon, and Koji Yatani. 2023. An Embroidery Touch Sensor With Layered Structure of Conductive and Nonconductive Threads. *IEEE Sensors Letters* 7, 6 (2023), 1–4. doi:10.1109/LSENS.2023.3277889

[11] Shahood uz Zaman, Xuyuan Tao, Cedric Cochrane, and Vladan Koncar. 2022. Smart E-Textile Systems: A Review for Healthcare Applications. *Electronics* 11, 1 (2022). doi:10.3390/electronics11010099

[12] Yaya Zhang, Jiyong Hu, Xiong Yan, and Huating Tu. 2021. The equivalent resistance model of double-layer embroidered conductive lines on nonwoven fabric. *Journal of Industrial Textiles* (2021), 8565S–8581S. arXiv:https://doi.org/10.1177/15280837211049481 doi:10.1177/15280837211049481